

EGRE 245 Engineering Programming

Spring 2026

Mark Schwitzerlett

Programming Assignment #2: Modular Sensor Data Analysis

1 Overview

In this assignment, write a C program that stores and analyzes sensor readings using a two-dimensional array and functions.

The program will read data for multiple sensors, compute statistics for each sensor, and display a formatted report.

Your program should

- Request the user to input readings for 4 sensors with 6 readings each.
- Store the readings in a two-dimensional array.
- Use functions to compute:
 - the average of a sensor's readings
 - the maximum reading
 - the minimum reading
- Use a function to print a formatted report.
- Output the average, maximum, and minimum for each sensor.

2 Code Documentation

The very first lines in all of your programming project files this semester should be comments in the class header file (copy from header.c).

In addition, the first output for all of your programming assignments should be the name of the assignment and your name, for example:

```
printf("Programming Assignment - Joe Student\n");
```

3 Sample Run

Your output should match the sample run below.

```
Programming Assignment #2 - Mark Schwitzerlett

Enter readings for Sensor 1
Reading 1: 2.1
Reading 2: 2.4
Reading 3: 2.2
Reading 4: 2.3
Reading 5: 2.5
Reading 6: 2.0

Enter readings for Sensor 2
Reading 1: 1.8
Reading 2: 2.4
Reading 3: 3.0
Reading 4: 2.9
Reading 5: 1.75
Reading 6: 2.25

Enter readings for Sensor 3
Reading 1: 2.2
Reading 2: 2.5
Reading 3: 2.6
Reading 4: 2.6
Reading 5: 2.4
Reading 6: 2.3

Enter readings for Sensor 4
Reading 1: 2.1
Reading 2: 2
Reading 3: .85
Reading 4: 1.25
Reading 5: 2.3
Reading 6: .9

Sensor   Average   Maximum   Minimum
1        2.25     2.50     2.00
2        2.35     3.00     1.75
3        2.43     2.60     2.20
4        1.57     2.30     0.85

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function.

All programming assignments this semester will be submitted in Gradescope (via Canvas).

Your file should be named:

`Assignment#2_FirstName_LastName.c`

Due date: This assignment is due at the end of your lab **Monday, April 20th or Tuesday, April 21.**

Be sure to save a copy of your .c file.